THE SKEW HOOK FORMULA OF NARUSE–OKADA DOES NOT EXTEND TO SIMPLE-SPECTRUM HEAPS

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Abstract. Let P be the coloured poset attached to a Weyl group element w for which the torus acts with simple spectrum, in the sense of [1, §5.4.2]. Every such P is a hook length poset, so its reverse-plane-partition series is a hook product; it need not be d-complete. We answer, negatively, the question of whether the skew hook formula of Naruse–Okada [3, Thm. 1.2] extends to this larger class. The obstruction is not the sum over excited diagrams but the hook function itself, and it is already visible at $F = \varnothing$, where the formula degenerates to the Peterson–Proctor product. We exhibit an eight-element witness in D_4: its true hook multiset is $\{1, 1, 3, 4, 4, 5, 6, 7\}$, and the recursion of [3, Def. 2.6] — which off the d-complete locus is a *family* of recursions, one per choice of d_k-interval — produces none of $\{1, 1, 3, 4, 4, 5, 6, 7\}$ under any of its choices, in either orientation convention. The verification is finite and small enough to check by hand: eight elements, at most eight choices. We emphasise what is *not* disproved: since P is a hook length poset, the $F = \varnothing$ case of the formula is true for P with the correct hook function, so the question of a corrected recursion remains open, and we state it.

1. The question, and the answer

Fix a simply-laced Dynkin diagram Γ with vertex set I and Weyl group W, a dominant weight λ, and $w \in W$. Following [1, §5.1], a reduced word for w has a *heap*: the poset on $\{1, \dots, \ell\}$ generated by the relations between positions carrying equal or adjacent colours, with elements coloured by $\pi(a) = i_a$. When w is λ-minuscule the heap is d-complete, and [3, Thm. 1.1] — the Peterson–Proctor hook-product formula — computes its reverse-plane-partition series. Dinkins, Karpov and Krylov observe [1, §5.4.2] that the relevant hypothesis is weaker than λ-minuscule: what is needed is that the torus act on each V_i with *simple spectrum*, and they ask for a combinatorial characterisation of such w. For every such w the resulting coloured poset $P = P(w)$ is conjectured, and in a companion manuscript proved, to satisfy

$$\sum_{\varphi \in \mathrm{rpp}(P)} q^{|\varphi|} \;=\; \prod_{u \in P} \frac{1}{1 - q^{\,h(u)}} \tag{1}$$

for a function $h : P \to \mathbb{Z}_{>0}$ — that is, P is a *hook length poset* in the sense of Ishikawa–Tagawa [2] — and the class of such P strictly contains the d-complete ones. Nothing below depends on that general statement: the present note needs (1) only for the single poset of §3, where it is verified directly in Lemma 3.2.

The question addressed here was put to the author by V. Krylov in October 2026.

Question 1.1 (Krylov)**.** Naruse and Okada prove a *skew* hook formula [3, Thm. 1.2]: for P d-complete and $F \subseteq P$ an order ideal, the series of reverse plane partitions on $P \setminus F$ is a finite sum over excited diagrams of F in P, with hook function given by [3, Def. 2.6]. Does this formula extend to the skew-heaps $P \setminus F$ arising in the simple-spectrum setting, where P is a hook length poset but need not be d-complete?

Theorem 1.2. *No. There is a heap P with eight elements, arising from a reduced word in D_4, which is a hook length poset, is not d-complete, and for which* (1) *fails for* every *hook function produced by* [3, Def. 2.6]*. Consequently the skew formula* [3, Thm. 1.2]*, read with its own hook function, is false for P already at $F = \varnothing$.*

The witness is Proposition 3.1; the two finite verifications are Lemma 3.2 and Proposition 4.1. Section 5 says precisely what the theorem does and does not rule out, and §6 states the question that replaces Question 1.1.

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2. The reduction to $F = \varnothing$

We use only the degenerate case of the skew formula. As [1, §5.3] note in quoting it: when $F = \varnothing$, so that $P \setminus F = P$, the set of excited diagrams consists of the empty diagram alone, the set of excited peaks is empty, and the formula reduces to (1). So if [3, Thm. 1.2] holds for P with hook function h, then (1) holds with that h.

It therefore suffices to show that no h permitted by [3, Def. 2.6] satisfies (1). The next lemma is what makes this a statement about *multisets*, and hence finitely checkable.

Lemma 2.1. *Let $(a_i)_{i\in A}$ and $(b_j)_{j\in B}$ be finite multisets of positive integers with $\prod_i (1-q^{a_i})^{-1} = \prod_j (1-q^{b_j})^{-1}$ in $\mathbb{Z}[[q]]$. Then the multisets coincide.*

Proof. Write c_d for the multiplicity of d in (a_i). Taking logarithms,

$$\log \prod_i \frac{1}{1-q^{a_i}} = \sum_{m\geq 1} \frac{1}{m} \sum_i q^{a_i m} = \sum_{N\geq 1} q^N \sum_{d\mid N} c_d \, \frac{d}{N},$$

so $N\,[q^N]\log(\cdot) = \sum_{d\mid N} d\,c_d$ for every $N \geq 1$. The series therefore determines $\sum_{d\mid N} d\,c_d$ for all N, and these determine $d\,c_d$, hence c_d, by induction on N (equivalently by Möbius inversion). The same holds for (b_j), and the two sets of data agree. □

Accordingly we call the multiset $\{h(u) : u \in P\}$ of any h satisfying (1) the *true hook multiset* of P; by Lemma 2.1 it is well defined, and it is an invariant of the series alone.

3. The witness

Proposition 3.1. *Let $\Gamma = D_4$ in the Bourbaki labelling, so that vertex 2 is trivalent with neighbours $1, 3, 4$, and let*

$$w \;=\; s_2\, s_1\, s_2\, s_3\, s_4\, s_2\, s_1\, s_3,$$

a reduced word of length 8. Its heap P has eight elements $p_1 \prec \cdots$ with colours and order

position	p_1	p_2	p_3	p_4	p_5	p_6	p_7	p_8
colour	2	1	2	3	4	2	1	3

$$p_1 \prec p_2 \prec p_3 \prec \{p_4, p_5\} \prec p_6 \prec \{p_7, p_8\},$$

with p_4, p_5 incomparable (colours $3, 4$ are non-adjacent) and p_7, p_8 incomparable (colours $1, 3$ are non-adjacent). Then P is not d-complete.

Proof. The order is the transitive closure of the heap relations, read off from the word: consecutive equal or adjacent colours are comparable, and the two displayed pairs carry non-adjacent colours. For the failure of d-completeness, no computation is needed: p_7 and p_8 are incomparable and have the common lower cover p_6, so they span a d_3^--interval, and Proctor's axiom at $k = 3$ requires such an interval to have a top, i.e. a common upper cover. Both p_7 and p_8 are maximal in P, so they have no upper bound at all. (The dual poset is not d-complete either, so the conclusion is independent of the orientation convention; see Remark 4.2.) □

Lemma 3.2. *The true hook multiset of the poset P of Proposition 3.1 is*

$$\{1, 1, 3, 4, 4, 5, 6, 7\}.$$

Proof. Both sides of (1) are computed independently and compared to order q^{26}. For the left-hand side, a reverse plane partition $\varphi : P \to \mathbb{Z}_{\geq 0}$ with $\varphi(x) \leq \varphi(y)$ for $x \prec y$ is equivalent to the decreasing chain of order filters $F_k = \{x : \varphi(x) \geq k\}$, with $|\varphi| = \sum_{k\geq 1} |F_k|$, so

$$\sum_{\varphi \in \mathrm{rpp}(P)} q^{|\varphi|} \;=\; 1 + \sum_{\varnothing \neq F} g(F), \qquad g(F) = \frac{q^{|F|}}{1-q^{|F|}} \Big(1 + \sum_{\varnothing \neq G \subsetneq F} g(G)\Big),$$

the second identity because repetitions $F_k = F_{k+1}$ are permitted. P has ten nonempty order filters, so this is a ten-term recursion. It gives

$$1 + 2q + 3q^2 + 5q^3 + 9q^4 + 14q^5 + 21q^6 + 31q^7 + 45q^8 + 63q^9 + 87q^{10} + 118q^{11} + \cdots,$$

which agrees with $\prod_{h\in\{1,1,3,4,4,5,6,7\}}(1-q^h)^{-1}$ in every coefficient up to q^{26}. By Lemma 2.1 the multiset is the only one that can occur. As a sanity check on the first two coefficients by hand: $[q^1]=2$ counts the two maximal elements p_7, p_8, and $[q^2]=3$ counts $\varphi = 2\cdot\mathbf{1}_{p_7}$, $2\cdot\mathbf{1}_{p_8}$ and $\mathbf{1}_{p_7}+\mathbf{1}_{p_8}$. □

4. Exhausting Definition 2.6

Off the d-complete locus an element of P may be the top of more than one d_k-interval, so [3, Def. 2.6] does not define a single function: it defines one for each choice of d_k-interval at each such element, together with the option of using the order-ideal count $\#\{v : v \preceq u\}$ there instead. We exhaust that family.

Proposition 4.1. *For P as in Proposition 3.1:*

(1) in the orientation of Proposition 3.1, three elements top a d_k-interval, giving eight choices and six distinct multisets,

$$\{1,2,3,4,4,5,6,6\},\ \{1,2,3,4,4,5,6,7\},\ \{1,2,3,4,4,5,7,7\},$$

$$\{1,2,3,4,4,6,6,6\},\ \{1,2,3,4,4,6,6,7\},\ \{1,2,3,4,4,6,7,7\};$$

(2) in the opposite orientation — the corner-as-maximum convention in which [3, Def. 2.6] *is stated — two elements top a d_k-interval, giving six choices and two distinct multisets,*

$$\{1,1,3,4,4,5,7,8\}, \qquad \{1,1,3,4,4,6,7,8\}.$$

In neither case does $\{1,1,3,4,4,5,6,7\}$ occur.

Proof. Finite enumeration; eight and six cases respectively. □

Proof of Theorem 1.2. Suppose [3, Thm. 1.2] held for P with hook function h given by [3, Def. 2.6]. Taking $F=\varnothing$ gives (1) for h, so by Lemma 2.1 the multiset of h is the true hook multiset of P, which by Lemma 3.2 is $\{1,1,3,4,4,5,6,7\}$. Proposition 4.1 shows no choice in the family defining h has that multiset, in either orientation. □

Remark 4.2. Both orientations are recorded because the hook function is pinned where the series is a hook product, while [3, Def. 2.6] is stated in the dual convention, and the correspondence between the two is exactly the kind of bookkeeping in which an error would hide. The conclusion does not depend on the choice.

Remark 4.3. The failure in the orientation of (i) is close: $\{1,2,3,4,4,5,6,7\}$ differs from the true multiset in one entry, 2 against 1. In the orientation of (ii) it is gross: the two largest hooks come out $7,8$ instead of $6,7$, so the product has the wrong behaviour at $q\to 1$ and no cancellation can repair it.

5. What is, and is not, disproved

It is worth being exact, because the negative statement is narrower than "the skew formula fails" and the residue is the interesting part.

(1) *What fails* is [3, Thm. 1.2] read with its own hook function, on hook length heaps that are not d-complete. The obstruction is the hook function, and it is present at $F=\varnothing$, before excited diagrams play any role.

(2) *What does not fail* is (1) itself. The poset P of Proposition 3.1 *is* a hook length poset, by Lemma 3.2; the $F=\varnothing$ case of the formula is true for P, with the hook function realising $\{1,1,3,4,4,5,6,7\}$. So nothing here suggests that the shape of [3, Thm. 1.2] is wrong for skew-heaps. What is wrong is the recipe for h.

(3) *Nothing is claimed about* skew $F\neq\varnothing$ beyond what (i) forces, nor about any formula whose hook function is defined otherwise, nor about d-complete P, where [3, Thm. 1.2] is a theorem and is not in question.

For context on how typical the witness is: over all reduced words of length at most 8 in D_4, among the heaps that are hook length posets, the d-complete ones reach their true hook multiset in all 336 cases, and the non-d-complete ones fail to reach it in all 36. So the dichotomy in this range is exactly d-completeness. All 36 failures have $|P|=8$; the witness of Proposition 3.1 is one of them, and 8 is the smallest size at which a failure occurs in this range.

6. The question that replaces it

Question 6.1. Every simple-spectrum heap P is a hook length poset, so it carries a hook function h_P satisfying (1), unique as a multiset by Lemma 2.1. Is there a recursion computing h_P — agreeing with [3, Def. 2.6] on the d-complete locus — for which the skew formula [3, Thm. 1.2] holds on all of this class?

Theorem 1.2 says that the recursion of [3, Def. 2.6] is not such a recursion. It leaves open that one exists, and item (ii) of §5 is weak evidence that it might: the $F = \varnothing$ case, which is the only case we have tested against, holds throughout the class once h is chosen correctly.

7. Reproduction

All claims above are finite checks. The computations are available with the rest of the project at https://github.com/jaideepsaipadhi/dkk-simple-spectrum; in the `skewheap` directory,

- `python3 certify2.py` computes the left-hand side of (1) for the witness by the filter recursion of Lemma 3.2, in both orientations, and compares with the hook product — from scratch, sharing no code with the rest of the project;
- `python3 orient.py` exhausts the [3, Def. 2.6] family in both orientations, printing every reachable multiset (Proposition 4.1);
- `python3 minwit.py 8 D4` searches for the smallest witness and recovers the one above;
- `python3 reach.py 8 11 D4` produces the 336 and 36 of §5.

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References

[1] H. Dinkins, I. Karpov and V. Krylov, *The quantum Hikita conjecture via quasimaps*, arXiv:2608.16746.
[2] M. Ishikawa and H. Tagawa, *Leaf posets and multivariate hook length property*, RIMS Kôkyûroku **1913** (2014), 67–80.
[3] H. Naruse and S. Okada, *Skew hook formula for d-complete posets via equivariant K-theory*, Algebr. Comb. **2** (2019), no. 4, 541–571.

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